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A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere with North and South America. The Earth is framed by a thin white rectangular border. The background is a deep black space filled with numerous small, distant stars.

COMP-2024 / COMP-2039
**Artificial Intelligence
Methods**

Lecture 11
Modelling and Simulation



Learning Outcomes

IDENTIFY

1. The Nature of Simulation
2. Why Simulate?
3. Systems, Models & Simulation
4. Model Design (Conceptual Modelling)
5. Classification of Simulation Models



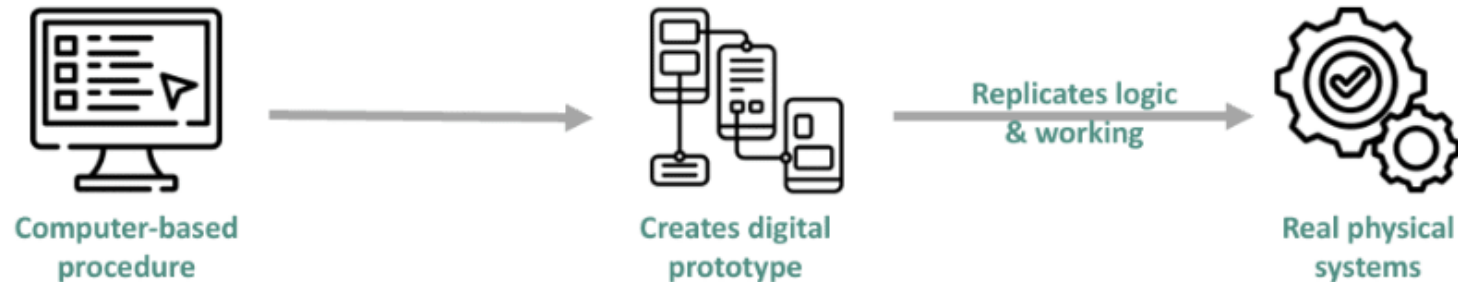
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Simulation

- **Simulation** is the *imitation* of the operation of a real-world process or system over time.
- It involves the creation of an *abstract model* that represents the key characteristics and behaviors of a selected physical or abstract system or process.
- The *model* is then used to *study* the system's operations, *predict* its behavior, and test different scenarios without disrupting the actual system.

Simulation Modelling



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The Nature of Simulation

Simulation

Imitate the operations of a facility or process, usually via computer

- What's being simulated is the (real world) **system (or process)**
- To study system, often make *assumptions/approximations*, both logical and mathematical, about how it works
- These assumptions form a **model** of the system
- If model structure is simple enough, could use *mathematical methods* to get *exact information* on questions of interest — *analytical solution*



The Nature of Simulation

- But most **complex** systems require models that are also complex (to be valid)
 - Must be studied via **simulation** — evaluate model numerically (mathematically) and collect data to estimate model characteristics
- **Example:** UNMC Super Capacitor Pilot Plant considers extending its plant
 1. *Build it* and see if it works out?
 2. *Simulate* current, expanded operations — could also investigate many other issues along the way, quickly and cheaply

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Why Simulate?

- It is possible with a simulation:
 - to *predict* system performance
 - to *compare* alternative system designs
 - to *determine* the *effects* of alternative policies on system performance
- **Advantages:** Simulation vs. Experimentation
 - Cost
 - Time (real time vs. virtual time)
 - Control of experimental conditions
 - If real system does not exist



Why Simulate?

- **Advantages:** Simulation vs. other modelling approaches:
 - *Modelling variability* (some other approaches could be adapted to account for variability but it often increases their complexity)
 - *Restrictive assumptions* (most of the other approaches require assumptions, e.g. queuing theory assumes particular distributions for arrival and service times, for many processes these distributions are not appropriate)
 - *Transparency* (more intuitive than a set of equations, an animated display of the system can be created, giving a non-expert greater understanding of, and confidence in, the model)
 - *Creating knowledge and understanding* (sometimes just building the model is enough)
 - *Visualisation, communication, interaction*

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Why Simulate?

- **Disadvantages** of simulation:
 - Expensive
 - Time consuming
 - Data hungry
 - Requires expertise (it's an art rather than a science)
 - Overconfidence (when interpreting the results from a simulation, consideration must be given to the validity of the underlying model and the assumption and simplifications that have been made!)



Applications

- Designing and analysing manufacturing systems
- Evaluating military weapons systems or their logistics requirements
- Determining hardware requirements or protocols for communications networks
- Determining hardware and **software requirements** for a computer system
- Designing and operating **transportation systems** such as airports, freeways, ports, and subways
- Evaluating designs for service organizations such as call centers, fast-food restaurants, hospitals, and post offices
- Reengineering of **business processes**
- Determining ordering policies for an inventory system
- Analysing financial or economic systems

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Modelling and Simulation Examples

Simulating Retail Management Practices

- Investigating the impact of human resource management practices on customer satisfaction

Modelling and Analysing Cargo Screening Processes

- Optimising the process flow of the cargo screening process
- Using simulation for cost-benefit analysis (risk assessment)

Future Energy Decision Making for Cities

- Modelling Energy Consumption Patterns
- Testing governmental intervention strategies

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System

System

A collection of **entities** (people, parts, messages, machines, servers, ...) that act and interact together toward some end (Schmidt and Taylor, 1970)

- In practice, depends on *objectives* of study
- Might limit the *boundaries* (physical and logical) of the system
- Judgment call: *level of detail* (e.g., what is an entity?)
- Usually assume a *time* element – dynamic system

State of a system

Collection of *variables* (attributes) and their values necessary to describe the system at that time

- Might depend on desired *objectives*, *output performance measures*
- Bank model: Could include number of busy tellers, time of arrival of each customer, etc.

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Types of Systems

Discrete

- State variables change instantaneously at separated points in time
- **Bank model:** State changes occur only when a customer arrives or departs

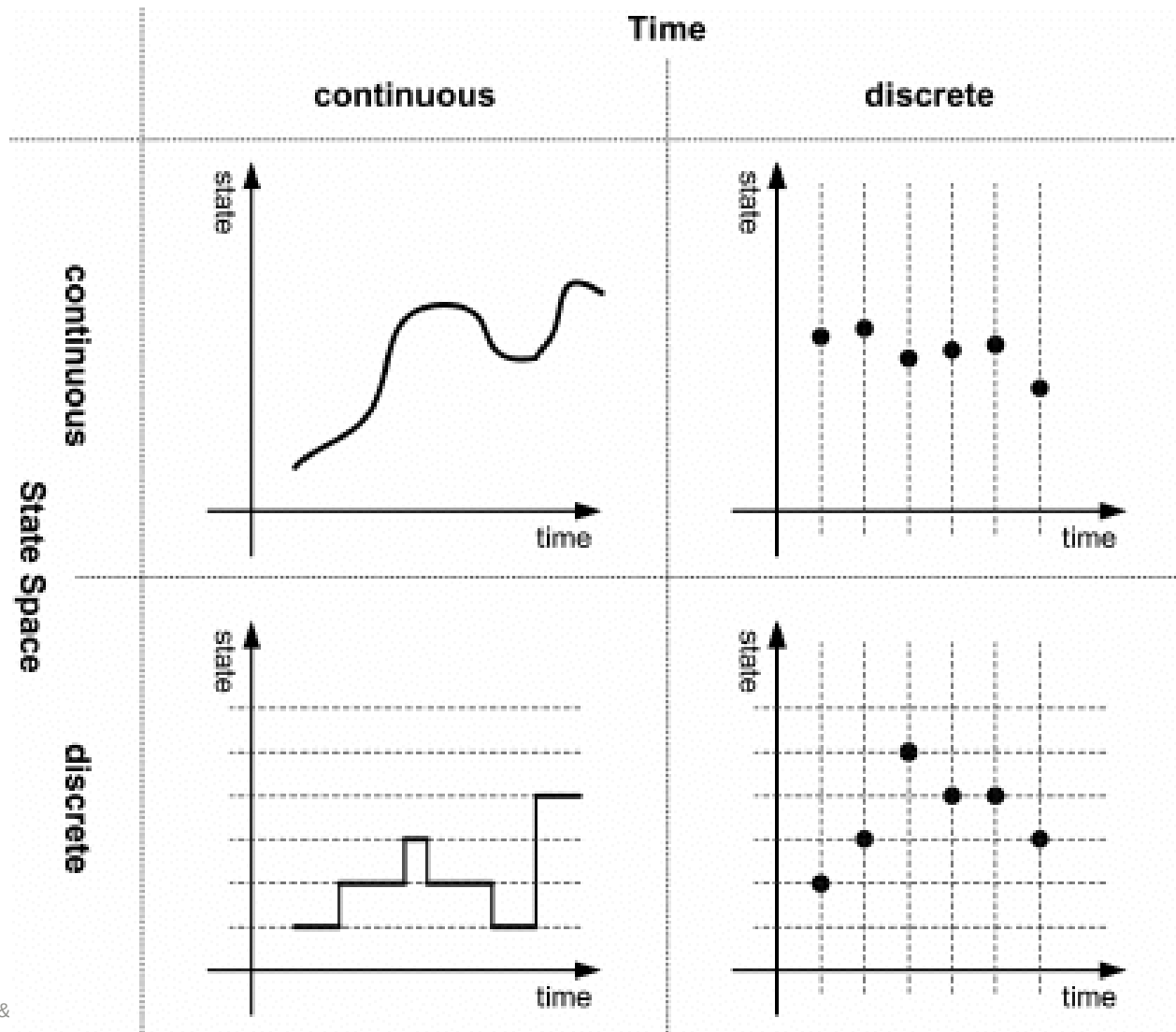
- State variables change continuously as a function of time
- **Airplane flight:** State variables like position, velocity change continuously

Continuous

Many systems are partly **discrete**, partly **continuous**



Discrete vs Continuous system

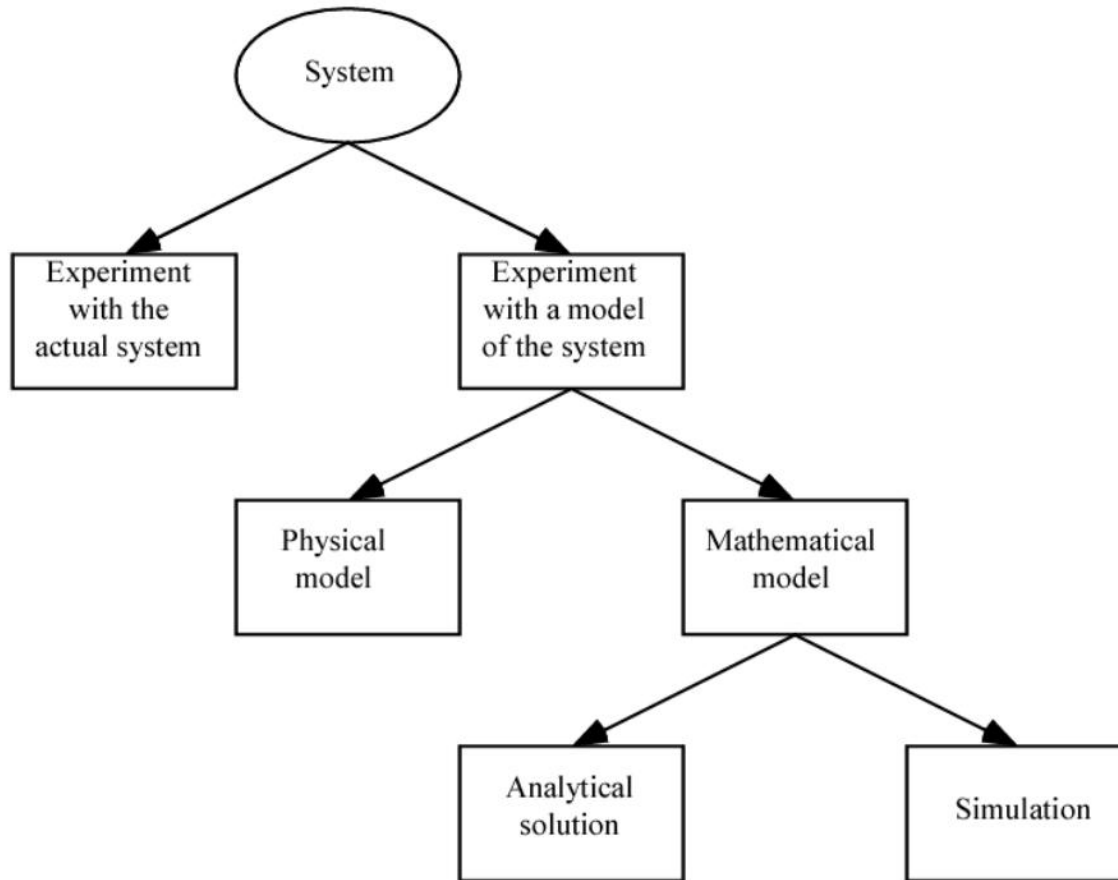


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Study a System

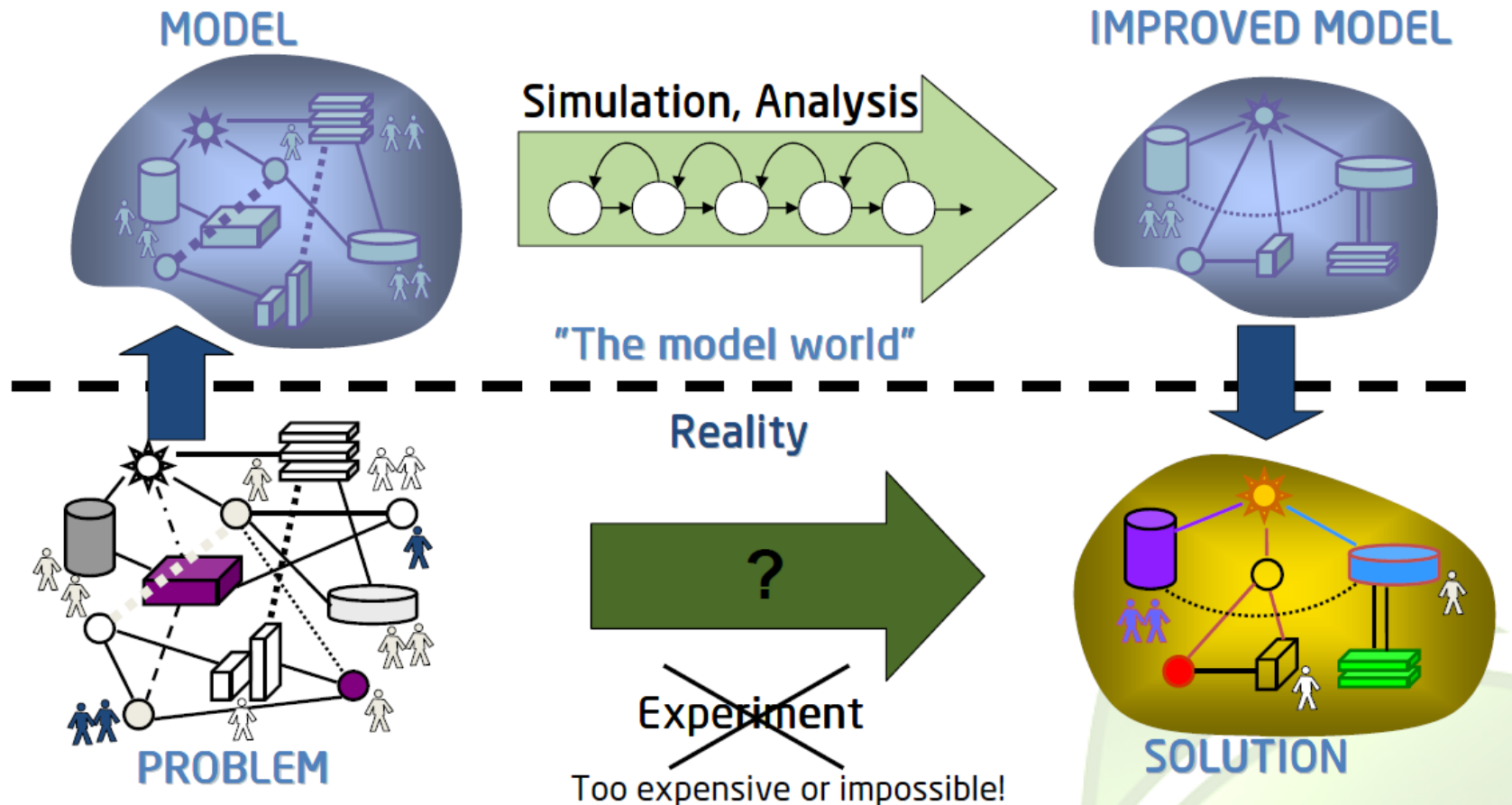
Ways to study a system



- **Simulation** is “method of last resort?” Maybe ...
- But with simulation there’s no need (or less need) to “look where the light is”



Simulation Modeling



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Simulation

- The process of *designing a **model*** of a real system and conducting **experiments** with this model for the purpose of understanding the behaviour of the system and /or evaluating various strategies for the operation of the system
- Uses a model to emulate the dynamic characteristics of a system

Simulation

Nature of simulation model use:

- Predict the performance of a system under a specific set of inputs
- Experimental approach to modelling (What-if analysis tool)



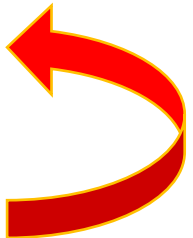
Models

Model

- Some form of **abstract** representation of a real system intended to promote understanding of the system it represents.
- A model is a *static representation* of the system
- Models can have many forms (e.g. mathematical equations, diagrams, physical mock-ups)

- Models give us a comprehensible representations of a system
 - Something to *think* about
 - Something to *communicate* about

Why Model?





Other alternatives

- Physical Experimentation
- Analytical Modeling
- Empirical Observation
- Expert Judgment
- Hybrid Approaches - Leverages the strengths of different methods to provide a more comprehensive understanding of the system.



Physical Experimentation

- Conducting **experiments** on the actual *physical system* or a scaled-down prototype.
- Provides direct *observation* and *measurement* of the system's behavior.
- Can be *time-consuming*, *costly*, and potentially risky or disruptive to the real system.



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Analytical Modeling

- Developing **mathematical models** and equations to describe the system's behavior.
- Relies on analytical techniques and closed-form solutions to predict the system's performance.
- Useful for *simple or well-understood systems*, but can be limited in capturing complex, dynamic, or nonlinear behaviors.

Find a closed form for the recursively defined sequence

$a_0 = 4$ $a_n = a_{n-1} - n$	$a_0 = 4$ $a_1 = (4) - 1$ $a_2 = ((4) - 1) - 2$ $a_3 = (((4) - 1) - 2) - 3$ $\vdots$ $a_n = ((((((4) - 1) - 2) - 3) \dots) - n) = 4 - (1 + 2 + \dots + n)$	$a_n = 4 - \frac{n(n+1)}{2}$
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Empirical Observation

- Collecting and analyzing **data** from the actual system through *field studies, surveys, or historical records*.
- Provides insights into the system's real-world behavior and performance.
- Can be time-consuming and limited in its ability to explore hypothetical scenarios or future conditions.



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Expert Judgment

- Relying on the knowledge and experience of **domain experts** to understand and predict the system's behavior.
- Can be *subjective* and may not capture the full complexity of the system.
- Useful for providing initial insights or complementing other analysis methods.



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What is a conceptual model?

Definition (*Robinson, 2004*):

The conceptual model is a **non-software specific description** of the simulation model that is to be developed, describing the objectives, inputs, outputs, content, assumptions and simplifications of the model.

Conceptual modelling is more an art than a science;
therefore it is difficult to define methods and procedures

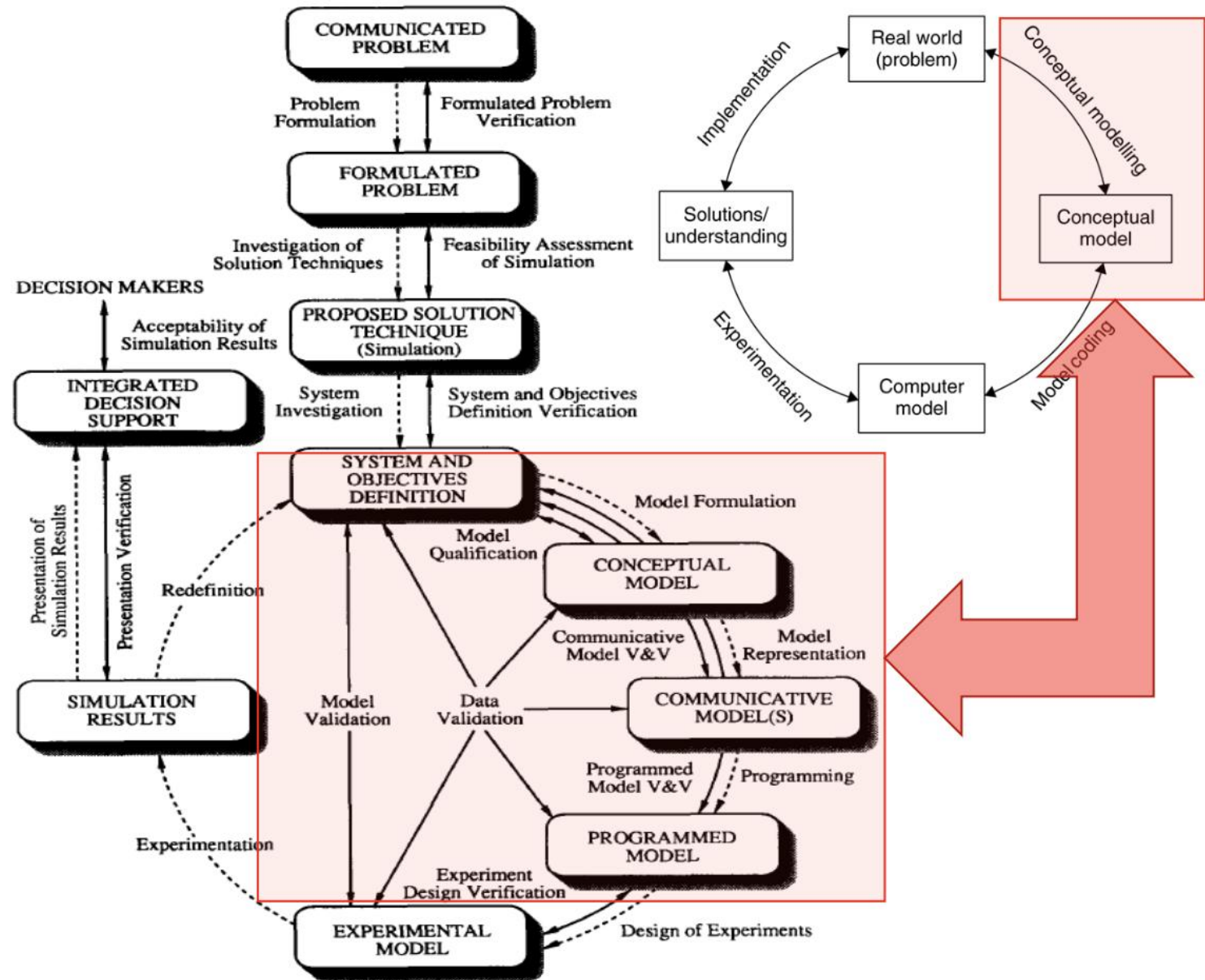


Conceptual modelling

- **Conceptual modelling** involves creating *abstract representations* of a domain or system, focusing on the key concepts, entities, and relationships within that domain, rather than the implementation details.
- Conceptual models are often *abstractions* of things in the real world, whether physical or social.



Oval symbols: Phases
Dashed arrows: Processes
Solid arrows: Credibility assessment stages



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Importance of model design (conceptual modelling)

- The modeller along with the clients determines the appropriate *scope* and *level of detail* to model, a process known as **conceptual modelling**
- Model design impacts all aspects of the study
- A high proportion of the benefits of a simulation study is obtained just from the development of the **conceptual model**
 - For the development of the conceptual model we often seek answers to questions that **have not** previously been asked
 - Effective conceptual modelling may even lead to the identification of a suitable solution **without** the need for any further simulation work

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What is a conceptual model? - Requirements

- **Validity:** A perception, on behalf of the *modeller*, that the conceptual model will lead to a simulation model that is *sufficiently accurate* for the purpose at hand
- **Credibility:** A perception, on behalf of the *clients*, that the conceptual model will lead to a simulation model that is *sufficiently accurate* for the purpose at hand
- **Utility:** A perception, on behalf of modeller and clients, that the conceptual model will lead to a simulation model that is *useful* as an aid to decision making within the specified context
- **Feasibility:** A perception, on behalf of modeller and clients, that the conceptual model will lead to a simulation model

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Take note of the terms



What is a conceptual model? – **Key Components**

Objectives

The purpose of the model

Inputs

Elements of the model that can be altered

Outputs

Measures to report the results from the simulation runs

Content

Components represented in the model and their interconnections

Assumptions

Uncertainties and beliefs about the real world to be incorporated into the model

Simplifications

Reduction of the complexity of the model



Model Complexity and Accuracy

Aim

Keep the model as *simple* as possible to meet the objectives of the simulation study



Advantages of simpler models

- They can be developed faster
- They are more flexible
- They require less data
- They run faster
- Results are easier to be interpreted

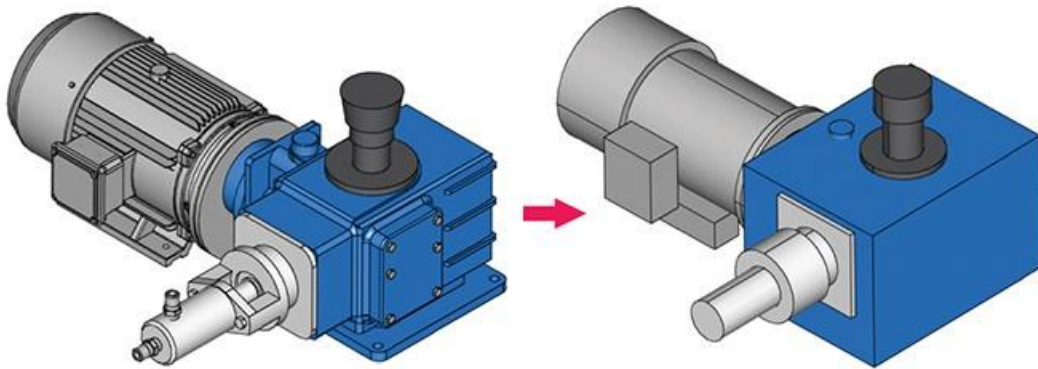
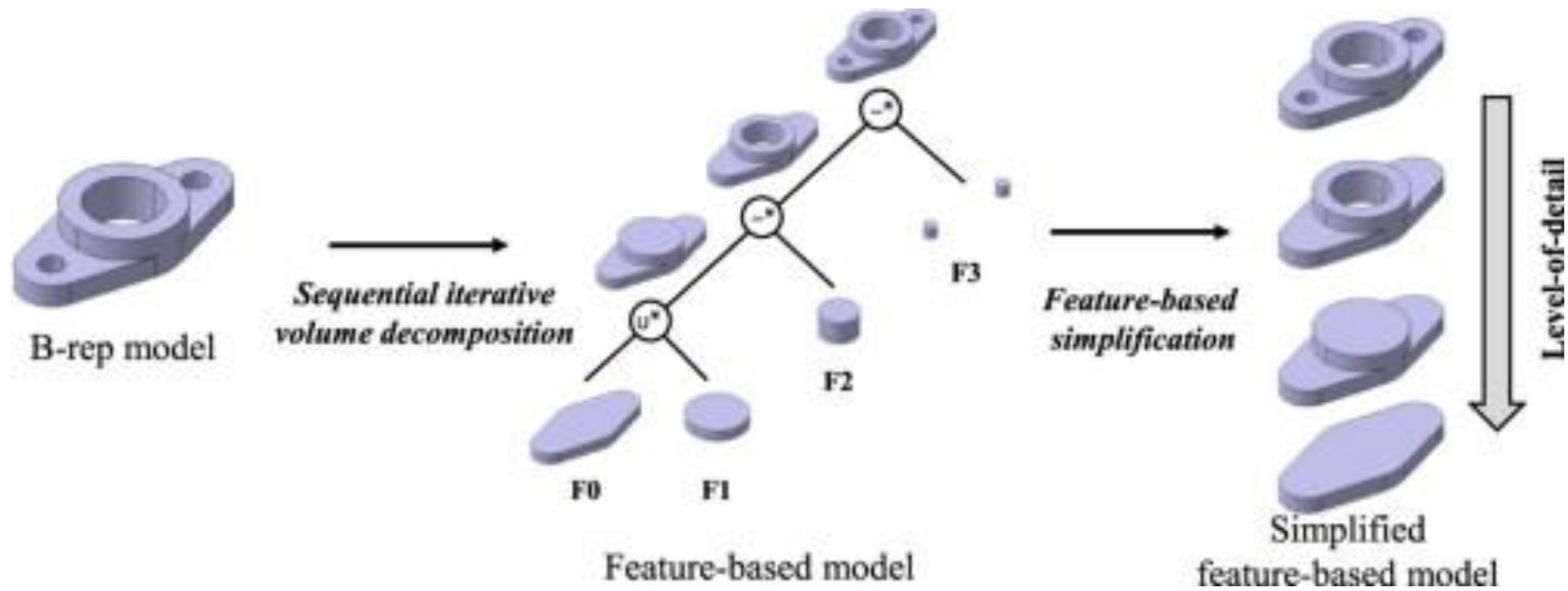


Methods of model simplification

- **Simplification** entails *reducing the scope and the level of detail* in a conceptual model
 - *Removing* components and interconnections that have little effect on model accuracy
 - Representing *more simple components and interconnections* while maintaining a satisfactory level of model accuracy
- Most effective approach to simplification is to *start with the simplest model possible* and gradually add to its scope and level of detail; once a point is reached in which the study objectives can be addressed, then no further details should be added



Model Simplification





Model Complexity and Accuracy

- 80/20 Rule

- 80 percent of accuracy is gained from only 20% of complexity; beyond this there is diminishing returns from increasing levels of complexity
- Increasing the complexity (scope and level of detail) too far might even lead to a less accurate model since the data and information are not available to support the detail being modelled



Methods of model simplification

Methods (scope or level of detail reduction?)

- Aggregation of model components
 - Black box modelling
 - Grouping entities
- Excluding components and details
- Replacing components with random variables
- Excluding infrequent events
- Reducing the rule set
- Splitting models

Over-simplification can make a model less transparent (overly abstract) and thereby reducing its credibility



Communicating / Representing the conceptual model



Communicating the Conceptual Model

- **Conceptual model** is part of the simulation project specification
- **Simulation Project Specification:**
 1. Background of the problem situation
 2. Objectives of the simulation study
 3. Expected benefits
 - 4. The conceptual model**
 5. Experimentation
 6. Data requirements
 7. Time scale and milestones
 8. Estimated costs



Classification of Simulation Models

- **Discrete:** Changes in the state of the system occur instantaneously at random points in time as a result of the occurrence of *discrete events*.
- **Continuous:** Changes of the state of the system occur *continuously* over time.

Discrete vs. Continuous

- **Static:** No attempts to model a time sequence of changes.
- **Dynamic:** Updating each entity at each occurring event.

Static vs. Dynamic

- **Deterministic:** *Rule based*
- **Stochastic:** Based on conditional *probabilities*

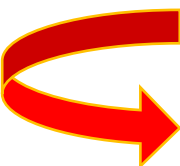
Deterministic vs. Stochastic

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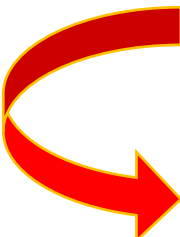
Communicating the Conceptual Model

Representing the conceptual model:



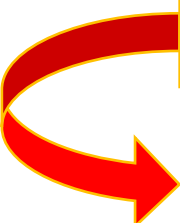
System Dynamics (SD)

Casual loop diagrams; Stock and flow diagrams



Discrete Event Simulation (DES)

Component list; process flow diagram; logic flow diagram; activity cycle diagram; combining Petri net and UML static structure diagrams (Pels and Goossenaerts, 2007)



Agent Based Simulation (ABS)

UML + AgentUML (class, component, sequence, deployment, state chart, use cases, and activity diagrams) (Bommel and Müller, 2008); coloured Petri nets (Jensen et al., 2007)

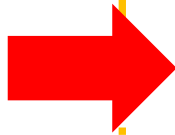


System Dynamics



System Dynamics (SD)

DEFINITION:



System Dynamics (SD) is a methodology and computer simulation modelling technique for framing, understanding, and discussing **dynamic problems**.

Mathematically, the basic structure of a formal system dynamics computer simulation model is a system of **coupled, nonlinear, first-order** differential (or integral) equations. Simulation of such systems is easily accomplished by *partitioning simulated time into discrete intervals of length dt and stepping the system through time one dt at a time.*

Example: Modeling the spread of a disease using *stocks* (infected population, recovered population) and *flows* (infection rate, recovery rate).



Applications of System Dynamics

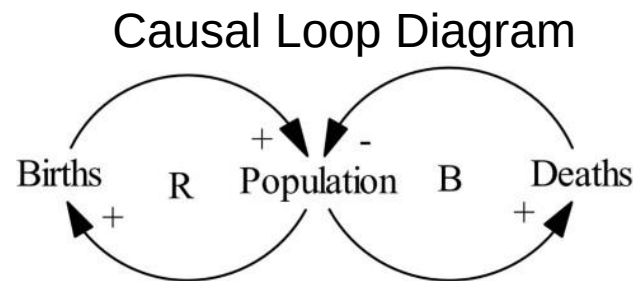
- Policy modeling and strategic decision-making (e.g., urban planning, resource management).
- Understanding long-term behaviors in economic, ecological / environmental, or social systems.
- Business process modeling for organizational change.
- Supply chains
- Healthcare systems



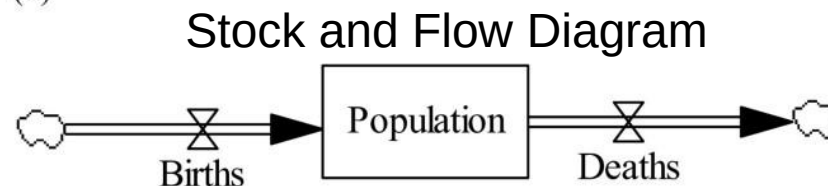
System Dynamics (SD)

- **System Dynamics** is a method for understanding, modeling, and analyzing complex systems that change over time. It focuses on the **structure** and **feedback loops** within a system to explain how behavior emerges.
- "Why is this system behaving the way it is—and how might it behave in the future?"

(a)



(b)



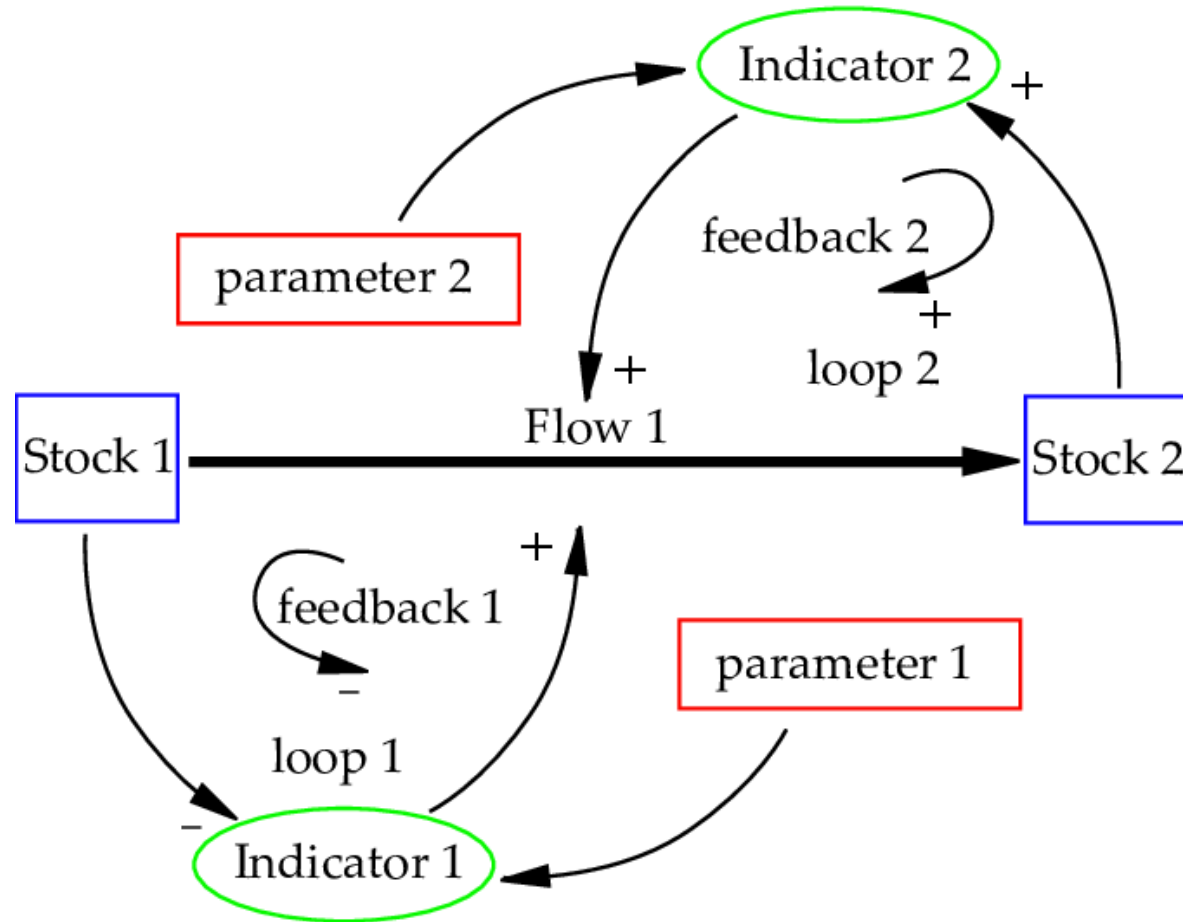


Key Concepts

- **Stocks:** Represent accumulations or levels (e.g., population, inventory).
- **Flows:** Represent rates of change into or out of stocks.
- **Feedback Loops:** Circular chains of cause-effect relationships (positive or reinforcing, and negative or balancing loops).
 - **Positive feedback:** Reinforces change (growth or decline).
 - **Negative feedback:** Stabilizes the system (resistance to change).
- **Time Delays:** Lag effects between actions and their outcomes.



System Dynamics



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Key Aspects of System Dynamics

- Causal Loop Diagram
- Stock and Flow Diagram
- Mathematical Modelling
- Computer Simulation: Software like Vensim, Stella, AnyLogic, or System Dynamics in Python (e.g., BPTK, PySD)

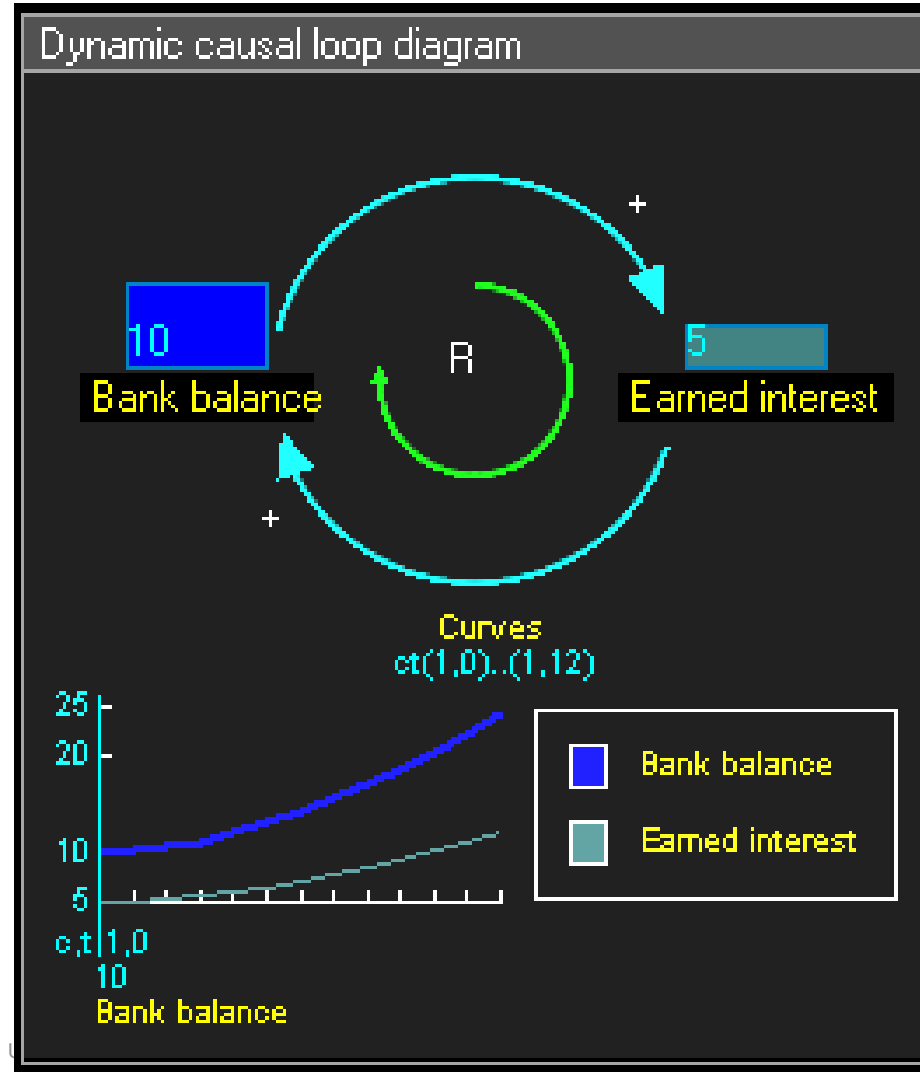


Causal Loop Diagram

- A **Causal Loop Diagram** shows how **variables in a system** are interrelated through **cause-and-effect relationships** and **feedback loops**.
- **Variables**: These are system elements (e.g., population, revenue, pollution).
- **Arrows ($\rightarrow$)**: Show the direction of causality.
- **Signs (+ / -)**:
 - + (**Positive**): A change in A causes B to change in the same direction.
 - - (**Negative**): A change in A causes B to change in the opposite direction.
- **Feedback Loops**:
 - **Reinforcing Loop (R)**: Amplifies change (growth or collapse).
 - **Balancing Loop (B)**: Pushes system toward equilibrium or stability.



Causal Loop Diagram

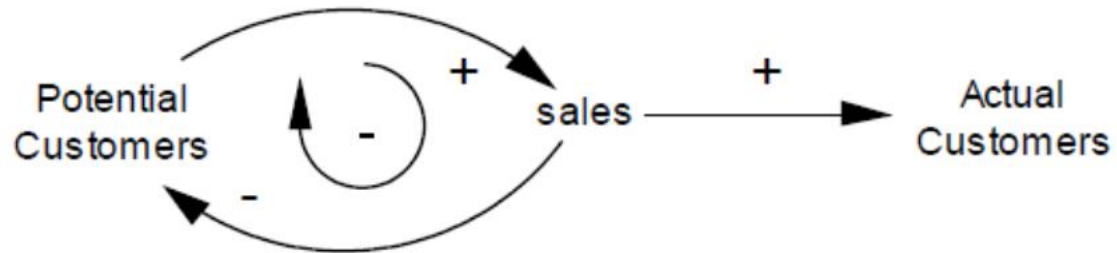


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Causal Loop → Stock and Flow

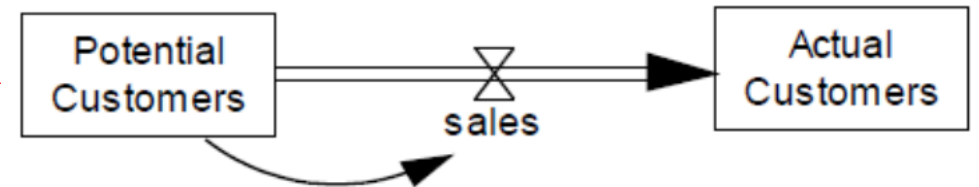
Example: Advertising for durable good



Causal loop diagram

Qualitative Analysis

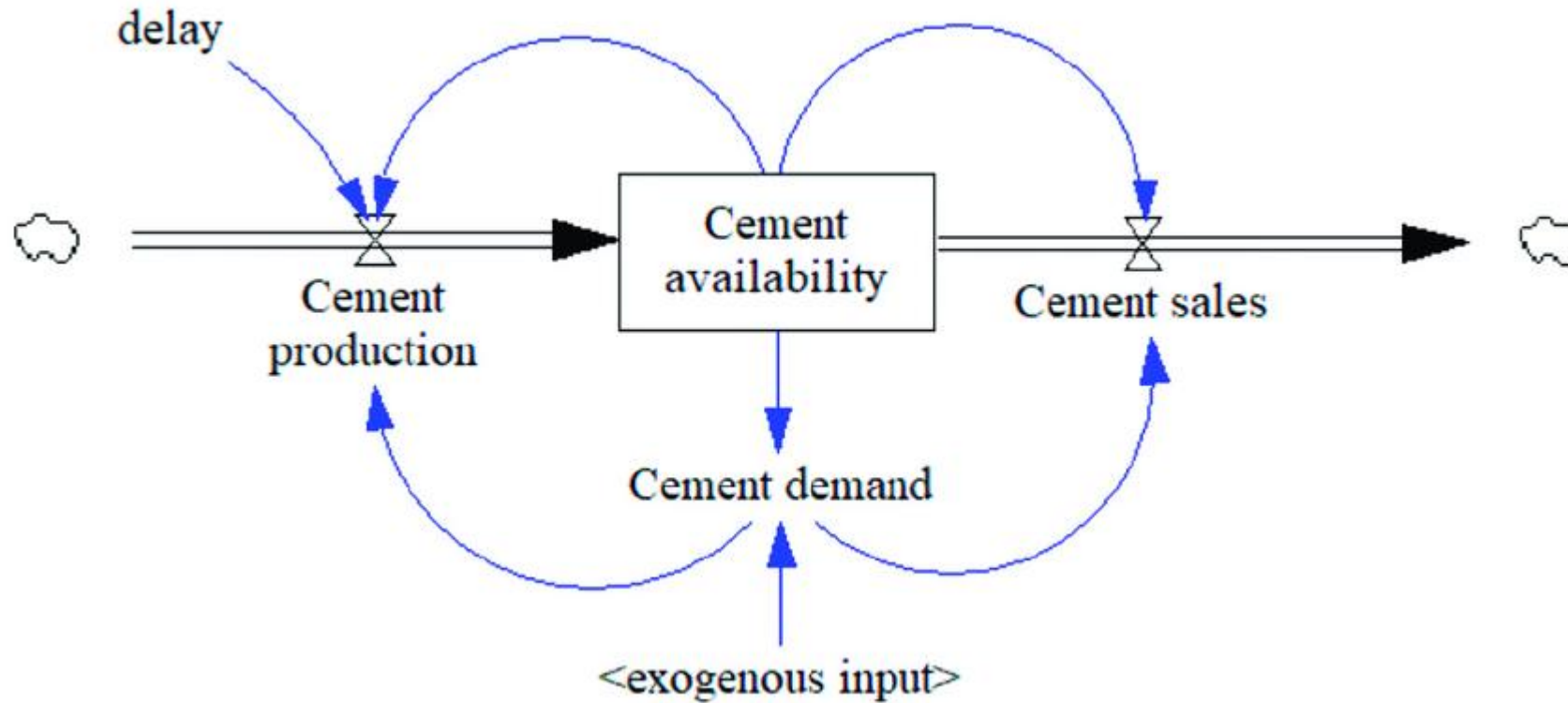
Quantitative Analysis



Stock and flow diagram



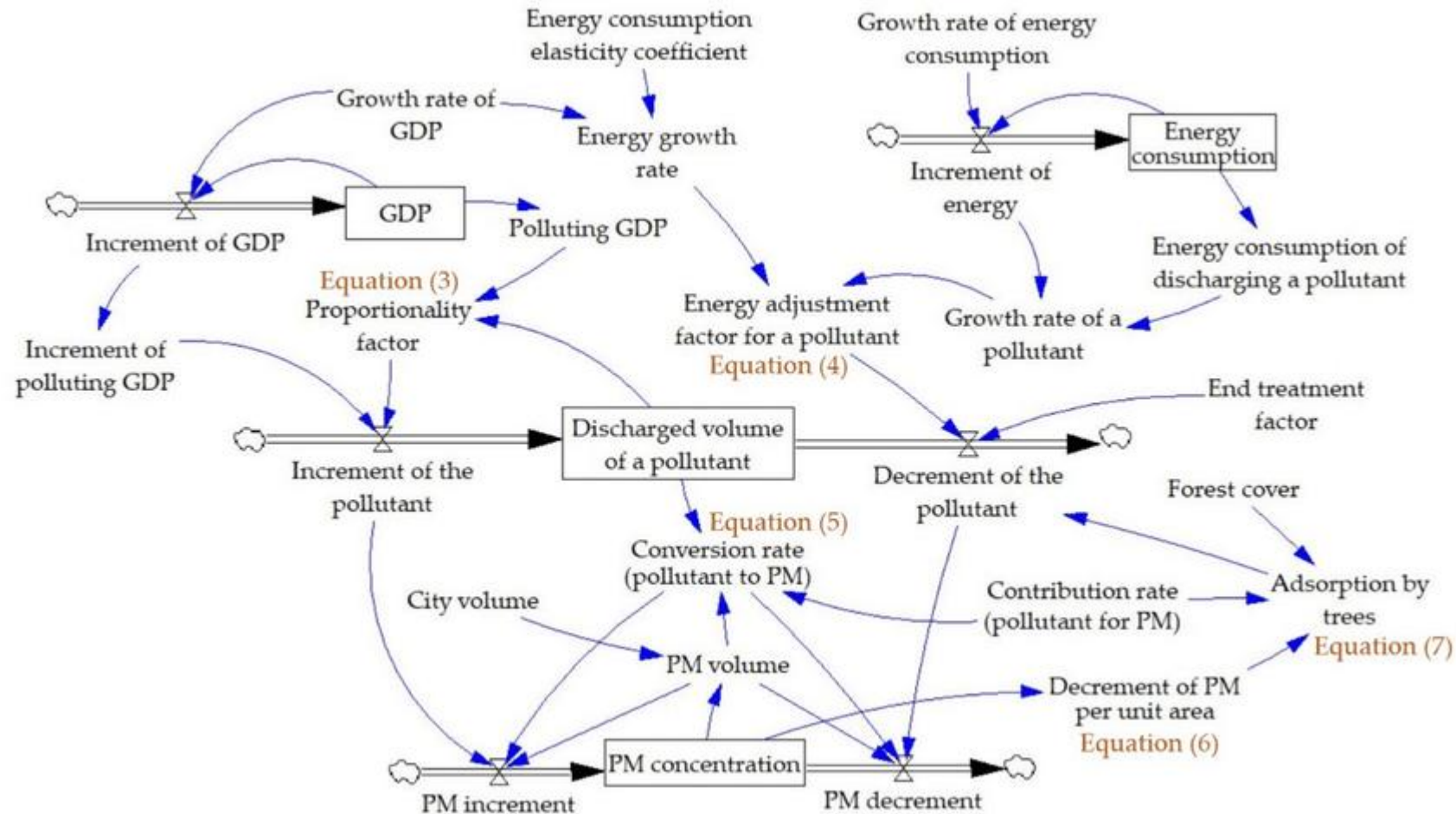
Example: Stock and Flow diagram



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Example: Stock and Flow diagram



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Stock and Flow Diagrams

- Shows relationships among variables which have the potential to change over time (like causal loop diagrams)
- **Distinguishes** between different types of variables (unlike causal loop diagrams)

Stock and flow diagram

Basic notation

Stock (level, accumulation, or state variable) {Symbol: **Box**}

- Accumulation of "something" over time
- Value of stock changes by accumulating or integrating flows
- Physical entities which can accumulate and move around (e.g. materials, personnel, capital equipment, orders, stocks of money)



Stock and Flow Diagrams

Flow (rate, activity, movement) {Symbol: **valve**}

- Flow or movement of the "something" from one stock to another
- The value of a flow is dependent on the stocks in a system along with exogenous influences

Information {Symbol: **curved arrow**}

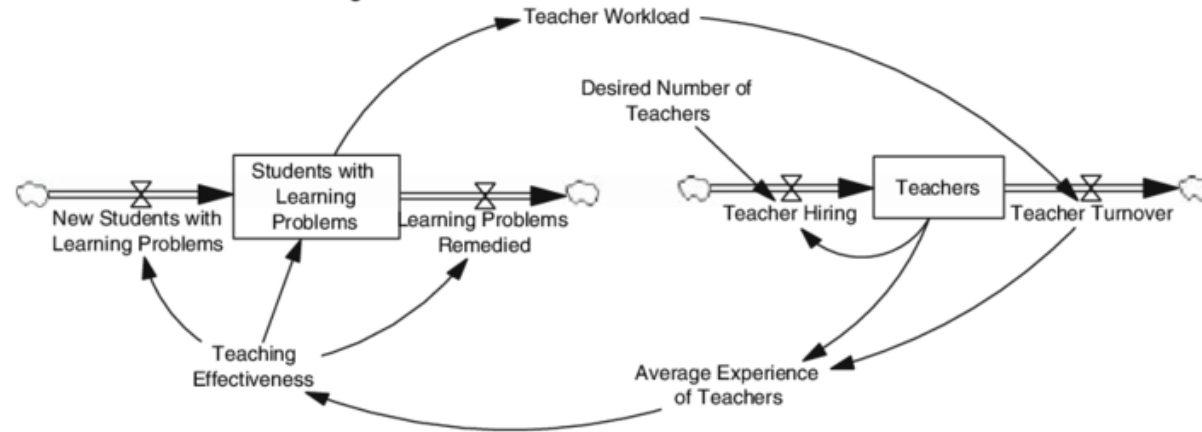
- Between a stock and a flow: Indicates that information about a stock influences a flow

Basic notation

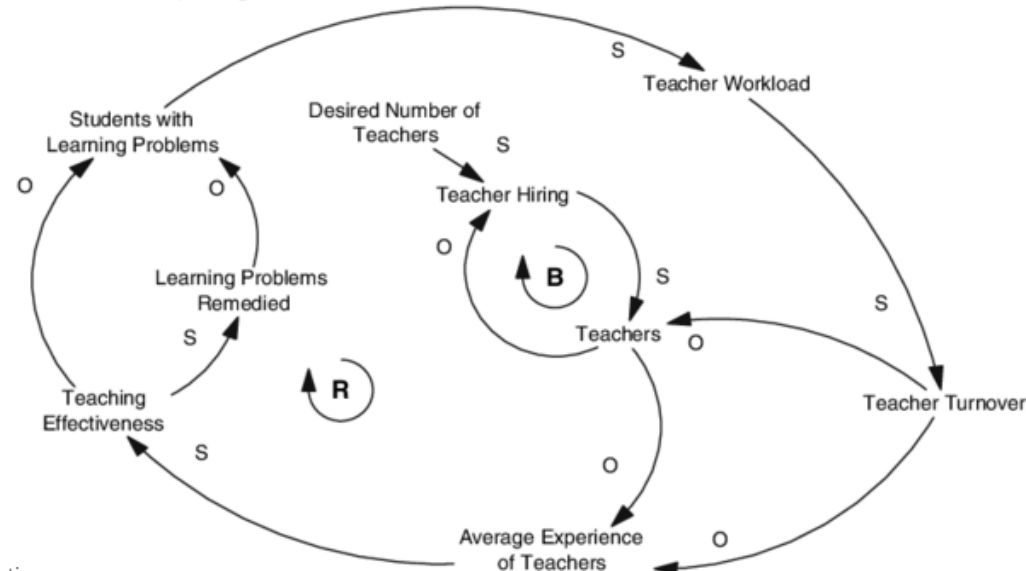


Causal Loop Diagrams vs Stock and Flow Diagram

A Stock and flow diagram



B Causal loop diagram





Causal Loop Diagrams vs Stock and Flow Diagram

Causal Loop Diagrams (CLDs)

- **Qualitative diagrams**
- They use arrows to show the direction of causality between variables, and positive/negative signs to indicate whether the relationship is reinforcing or balancing.
- Identify the underlying feedback structures that drive system behavior, but do not quantify the strength or timing of those relationships.

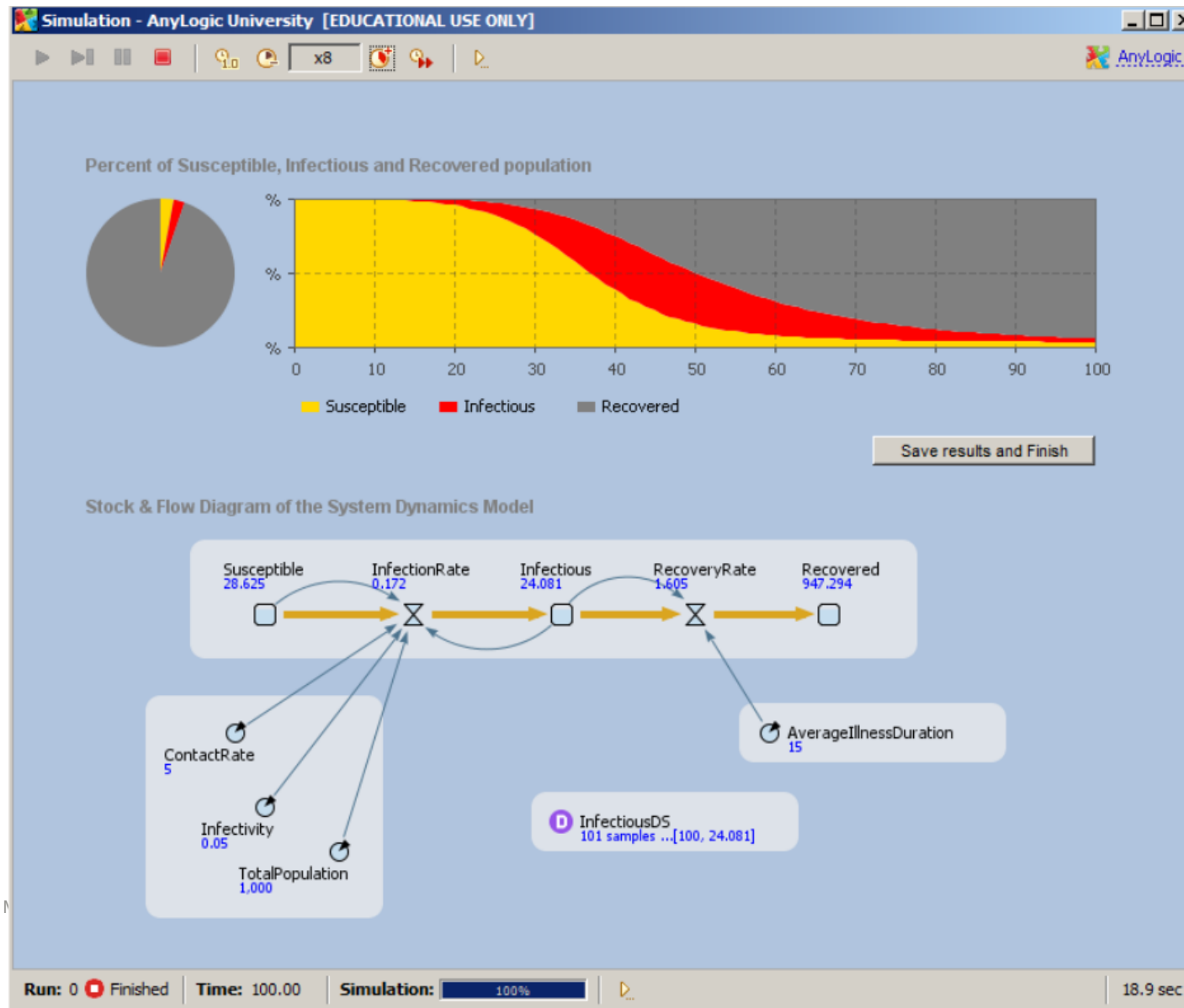
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Stock and Flow Diagrams (SFDs)

- **Quantitative, computational models**
- They represent the system as a set of *stocks* (accumulations) connected by *flows* (rates of change), with auxiliary variables and information links.
- Allow for the translation of conceptual understanding into a formal simulation model that can be used to test hypotheses and explore potential future scenarios.



System Dynamics Computer Simulation



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Learn Stock and Flow Diagram

- <https://thesystemsthinker.com/step-by-step-stocks-and-flows-improving-the-rigor-of-your-thinking/>
- <https://systemic2016.wordpress.com/system-dynamics-stock-and-flow-modelling/>
- <https://www.cs.sjsu.edu/faculty/pearce/modules/lectures/abs/sd/sysdynamics.htm>
- <https://www.youtube.com/watch?v=jehYN3yULHo>



How to build SD Simulation Models

Conceptualisation

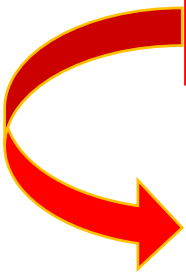
1. Define the *purpose* of the model
2. Define the model *boundaries* and identify key *variables*
3. Describe the *behaviour* of the key variables
4. Diagram the basic *mechanisms* (feedback loops) of the system

Formulation

1. Convert diagrams to *stock and flow equations*
2. Estimate and select *parameter values*
3. Create the *simulation model*

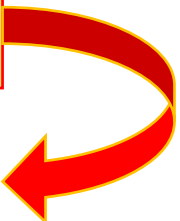


How to build SD Simulation Models



Testing

1. Test the dynamic hypothesis (the potential explanation of how structure is causing observed behaviour)
2. Test model behaviour and sensitivity to perturbations



Implementation

1. Test model's responses to different policies
2. Translate study insight to an accessible form



Discrete-Event Simulation



Discrete-Event Simulation

- **Discrete-event simulation:** Modelling of a system as it evolves over time by a representation where the state variables **change** instantaneously at separated (discrete) *points in time*
 - More precisely, state can change at only a countable number of (discrete) points in time
 - These points in time are when events occur
- **Event:** Instance of time in which a significant state change occurs
- Examples include queuing systems, manufacturing processes, transportation networks, etc. - systems where individual entities (customers, parts, vehicles, etc.) move through a series of discrete steps or activities.



Discrete-event Simulation vs System Dynamics

Discrete-event Simulation

- Focus on modeling the **individual entities**, **activities**, and **events** that occur within a system.
- Employ probabilistic and logical rules to govern the occurrence of events.
- Suited for operational optimization and resource planning.

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System Dynamics

- Focus on modeling the **continuous feedback** structures and **flows** that drive system behavior over time.
- Use differential/integral equations
- Suited for exploring how system structure gives rise to dynamic behavior



Discrete-event Simulation

1. **Initialize** the system state and the simulation clock.
2. **Schedule** future events (e.g., next arrival, next departure).
3. **Jump** to the time of the next scheduled event.
4. **Update** system state according to the event.
5. **Repeat** until a stopping condition is met (e.g., simulated 24 hours).



Discrete-event Simulation

- **Discrete:** Time doesn't flow continuously — it jumps from event to event (e.g., a customer arriving, a machine breaking down).
- **Events:** Instantaneous occurrences that change the state of the system (e.g., "Customer arrives", "Service begins").
- **Entities:** Objects moving through the system (e.g., people, products, data packets).
- **Resources:** Things that entities use (e.g., servers, machines, doctors).
- **Queues:** Where entities wait for resources (e.g., checkout lines, waiting rooms).
- **Simulation Clock:** Tracks the current time in the simulation.

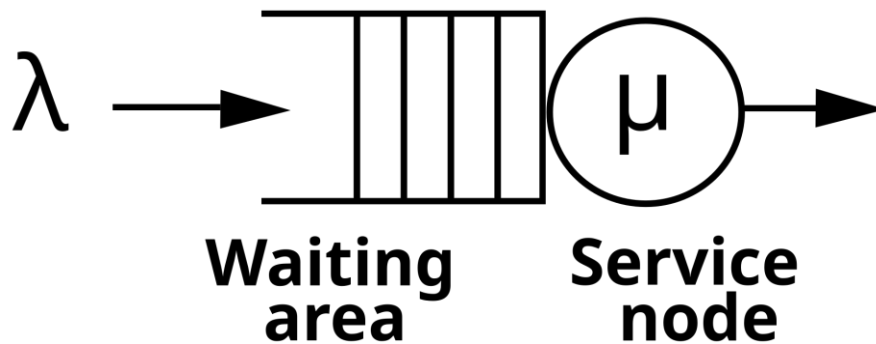
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Communicating the Conceptual Model

DES Example: M/M/1 Queue

- Represents a single server that has **unlimited** queue capacity and calling population, both arrivals and service are Poisson (or random) processes, meaning the statistical distribution of both the inter-arrival times and the service times follow the exponential distribution.





Communicating the Conceptual Model

DES Example: M/M/1 Queue

- Represents a single server that has **unlimited** queue capacity and calling population, both arrivals and service are Poisson (or random) processes, meaning the statistical distribution of both the inter-arrival times and the service times follow the exponential distribution.

Component	Detail
Customers	Time between arrivals (distributions)
Queue	Capacity
Service Desk	Service time (distributions)



Component list



M/M/1 queue

- **M/M/1** refers to a specific type of queueing model in queueing theory.
 - **M**: Markovian (or Memoryless) arrival process. This means the interarrival times of customers follow an exponential distribution, which is characteristic of a Poisson process.
 - **M**: Markovian (or Memoryless) service process. This means the service times are also exponentially distributed.
 - **1**: There is 1 server in the system.
- An extension of this model with more than one server is the M/M/c queue.

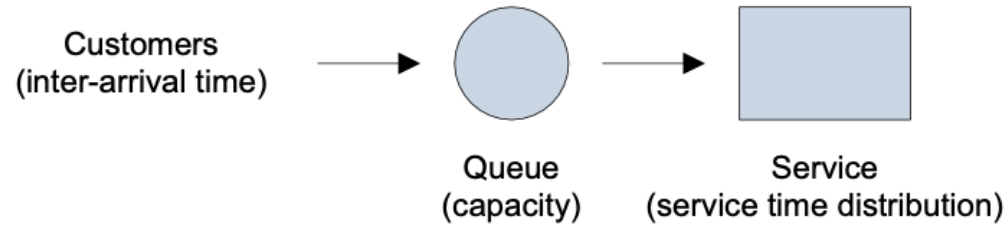


Applications of Discrete-event Simulation

- Operations and logistics (e.g., optimizing manufacturing, supply chains).
- Service industry modeling (e.g., customer service queues, hospital operations).
- Event-driven systems analysis (e.g., traffic flow, IT systems).

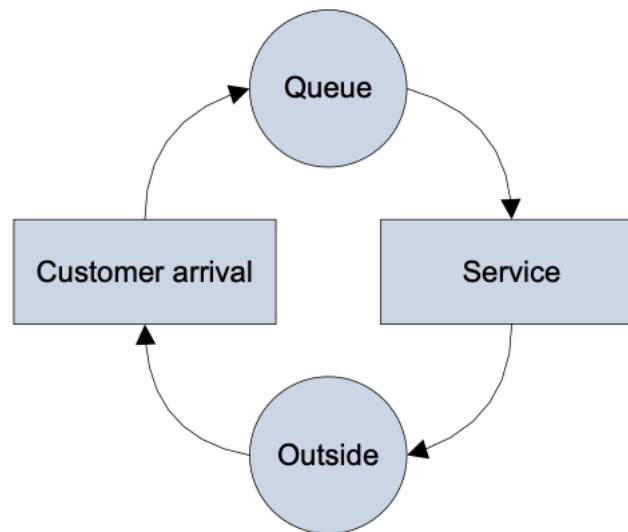


How to represent the Conceptual Model?

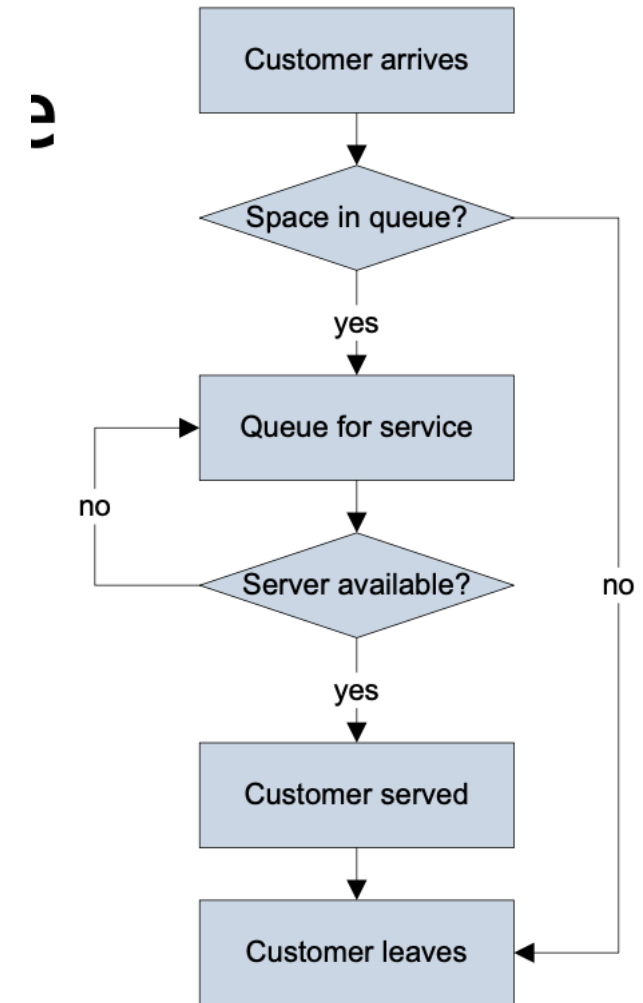


Process flow diagram

DES Example: M/M/1 Queue



Activity cycle diagram



Logic flow diagram

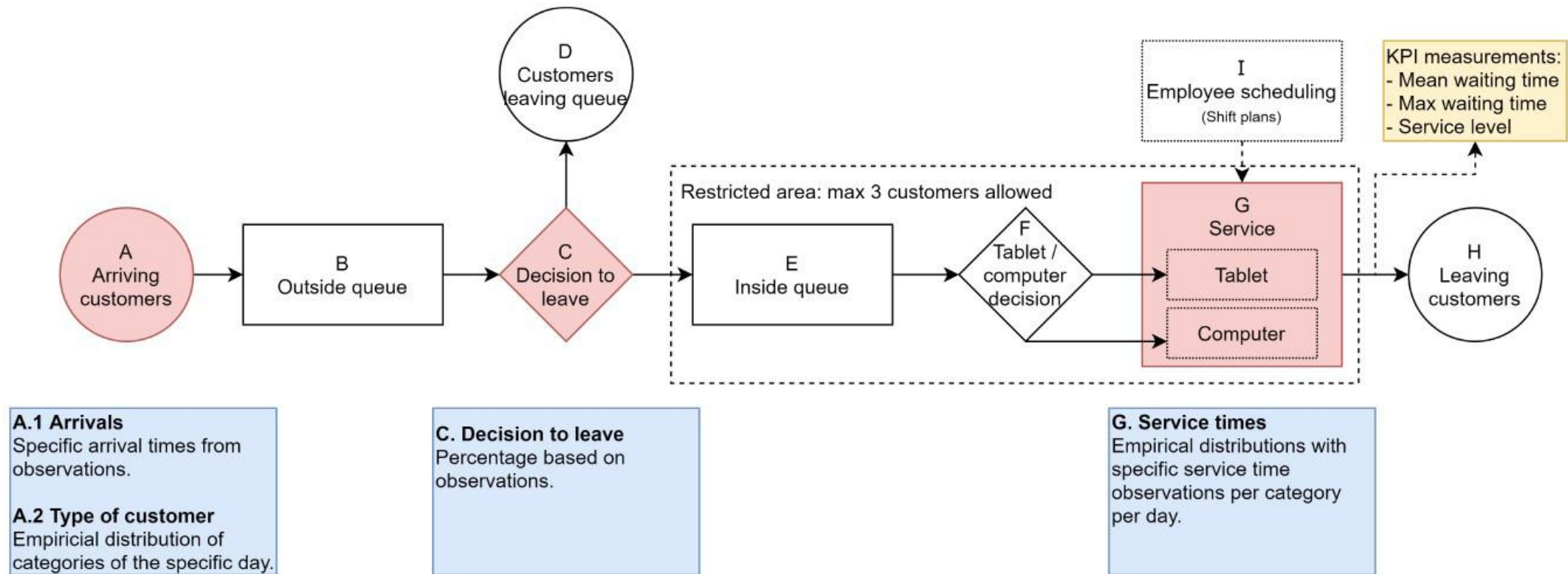


Example: Bank Queue Simulation

- **Entities:** Customers
- **Events:** Arrival, start service, end service
- **Resources:** Tellers
- **Queues:** Waiting line at the counter
- **Simulation tracks:**
 - How long customers wait
 - How busy the tellers are
 - How many customers leave unserved



Example: Bank Queue Simulation

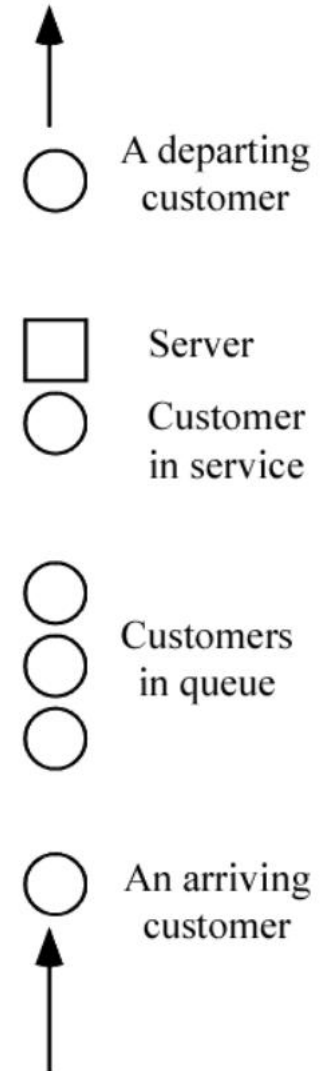


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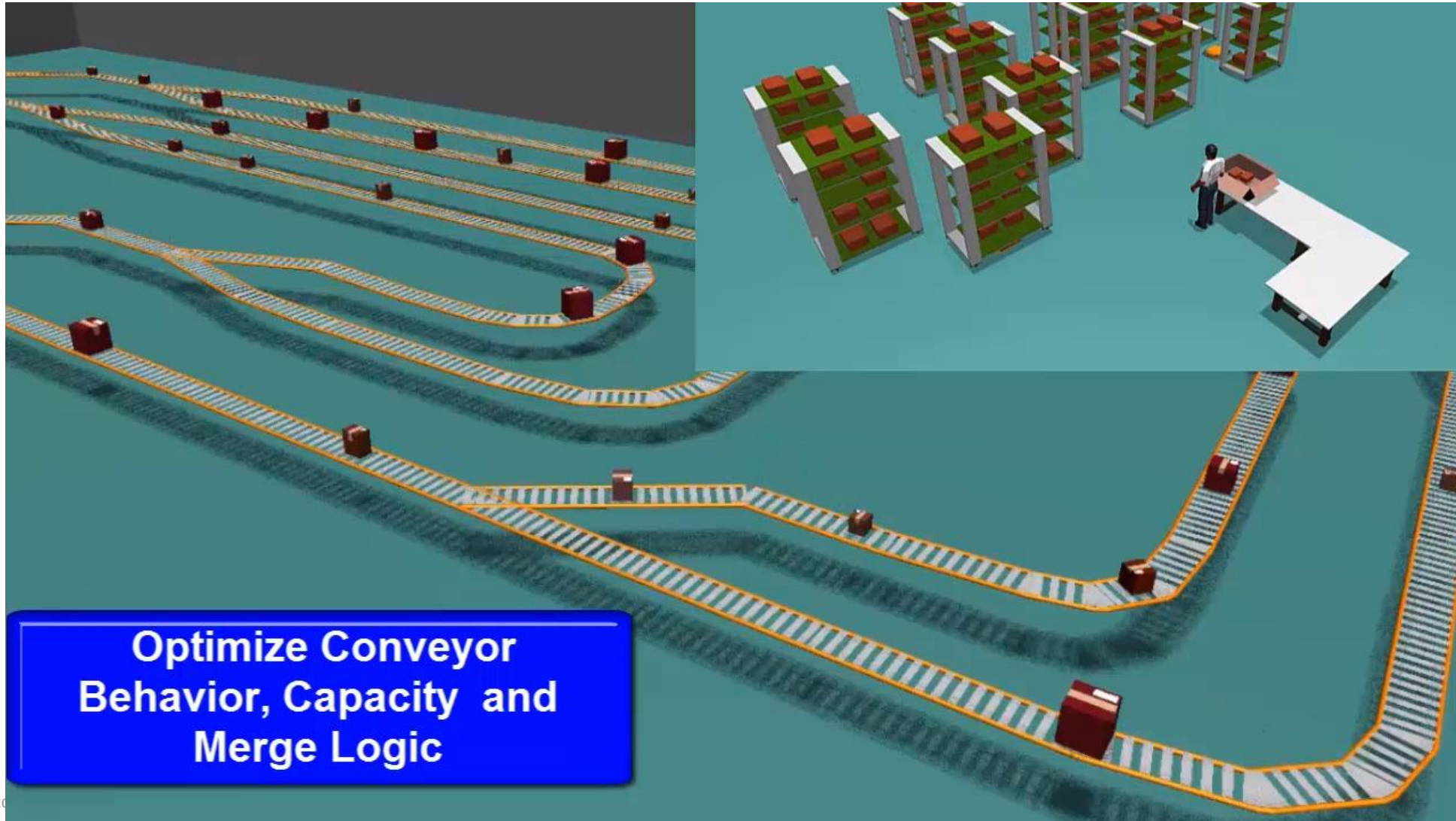
Example – Single-server queue

- Estimate expected average delay in queue (line, not service)
- **State variables**
 - Status of server (idle, busy) – needed to decide what to do with an arrival
 - Current length of the queue – to know where to store an arrival that must wait in line
 - Time of arrival of each customer now in queue – needed to compute time in queue when service starts
- **Events**
 - Arrival of a new customer
 - Service completion (and departure) of a customer
 - Maybe – end-simulation event (a “fake” event) – whether this is an event depends on how simulation terminates (a modelling decision)





Example: Conveyor System Discrete Event Simulation



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Conveyor System Discrete Event Simulation

■ State Variables:

- Number of parts on the conveyor
- Position of each part along the conveyor
- Status of the conveyor (running, stopped, etc.)
- Availability of upstream and downstream resources (e.g. loading stations, unloading stations)

■ Key Events:

- Part arrival at the conveyor input
- Part reaches loading station
- Part loading onto the conveyor
- Part movement along the conveyor
- Part reaches unloading station
- Part unloading from the conveyor
- Conveyor starts/stops
- Conveyor breakdown and repair



Conveyor System Discrete Event Simulation

- The discrete-event simulation would model the timeline of these events and how they impact the state of the system. For example:
 1. A part arrives at the conveyor input, joining the queue waiting to be loaded.
 2. When the loading station becomes available, the part is loaded onto the conveyor.
 3. The part then moves along the conveyor at a certain speed, updating its position over time.
 4. When the part reaches the unloading station, it is removed from the conveyor.
 5. If the downstream unloading station is blocked, the part may have to wait on the conveyor, impacting the queue and throughput.
 6. Breakdowns or stops in the conveyor operation would be modeled as discrete events that change the state of the system.

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DES Software Simulation

- SimPy (Python)
- Arena
- AnyLogic
- FlexSim
- Simul8

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Agent-Based Modelling (ABM)



Agent-Based Modelling (ABM)

- In **Agent-Based Modelling (ABM)**, a system is modelled as a collection of **autonomous decision-making entities** called **agents**. Each agent individually assesses its situation and makes decisions on the basis of a set of rules.
- ABM is a **mindset** more than a technology. The ABM mindset consists of describing a system from the perspective of its *constituent units*. [Bonabeau, 2002]
- ABM is well suited to modelling systems with **heterogeneous, autonomous and pro-active actors**, such as human-centred systems.
- Example: Simulating the spread of ideas in a social network with agents representing individuals who adopt ideas based on peer influence.

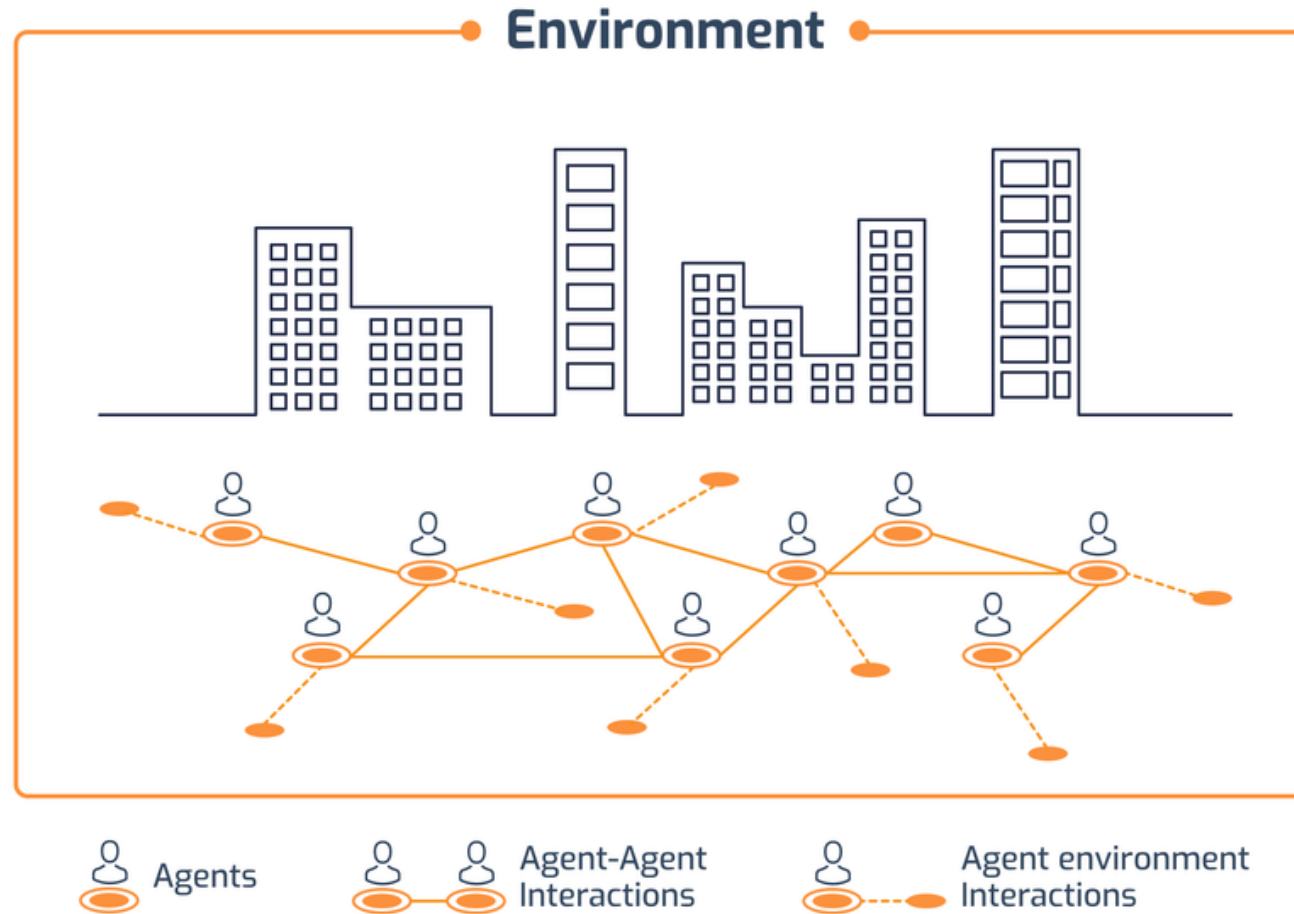


Key Concepts

- **Agents:** Individual entities with defined behaviors and decision-making capabilities.
- **Environment:** The space where agents interact and are influenced by local conditions.
- **Rules:** Guidelines for agent behavior and interactions (e.g., movement, decision-making).
- **Emergent Behavior:** System-level phenomena arising from agent interactions (e.g., traffic jams, market trends).



Agent-Based Modelling



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Agent-Based Modelling



What do we mean by "agent"?

- Agents are objects with **attitude!**

Discrete entities

- With their own *goals* and behaviours
- With their own thread of control

Autonomous

- Capable to adapt
- Capable to modify their behaviour

Proactive

- *Actions* depending on motivations generated from their internal state

Properties

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Agent-Based Modelling

- The agents can represent individuals, households, organisations, companies, nations, ... depending on the application.
- ABMs are essentially **decentralised**; there is no place where global system behaviour (dynamics) would be defined.
- Instead, the individual agents *interact* with each other and their environment to produce complex **collective behaviour patterns**.



Agent-Based Modelling

Benefits of AMB

- ABM provides a natural description of a system
- ABM captures **emergent phenomena**



Emergence

- **Emergent phenomena** result from the interactions of individual entities. The whole is more than the sum of its parts because of the interactions between the parts.
- An emergent phenomenon can have properties that are decoupled from the properties of the part.
- Example: Traffic Jam Dynamics

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Agent-Based Modelling

- When to use ABM? [Siebers et al. 2010]
 - When the problem has a **natural representation** as agents – when the goal is modelling the behaviours of individuals in a diverse population
 - When agents have **relationships** with other agents, especially dynamic relationships -agent relationships form and dissipate, e.g., structured contact, social networks
 - When it is important that individual agents have **spatial** or **geo-spatial** aspects to their behaviours (e.g. agents move over a landscape)
 - When it is important that agents learn or **adapt**, or populations adapt
 - When agents engage in **strategic behaviour**, and anticipate other agents' reactions when making their decisions



Agent-Based Modelling

- Many different developments have been going on under the slogan of **Agent Based Simulation** in very different disciplines, e.g. Robotics, Artificial Intelligence, Complexity Science, Economics, Social Science
- Each discipline has its own understanding of what constitutes an agents and a multi agent system

A word of caution

2

Main Paradigms

Multi-agent decision systems

- Usually embedded agents or a simulation of embedded agents
- Focus is on **decision making**

Multi-agent simulation systems

- The multi-agent system is used as a model to simulate some real-world domain and **recreate** some real world phenomena

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Multi-agent simulation systems

- **Multi-agent simulation systems** refer to a class of simulation models that incorporate multiple autonomous agents interacting within a shared environment. These systems are designed to study the *emergent behavior* and dynamics that arise from the *collective actions* and decisions of the individual agents.
- The collective behavior and patterns observed in a multi-agent simulation system emerge from the *local interactions* and *decisions* of the individual agents.
- This *emergent behavior* is often complex, non-linear, and difficult to predict based on the individual agent behaviors alone.



Agent-Based Simulation – MORE Examples

Field	Application Examples
Social Science	Insect societies, group dynamics in fights, growth and decline of ancient societies, group learning, spread of epidemics, civil disobedience
Economics	Stock market, self organising markets, trade networks, consumer behaviour, deregulated electric power markets
Ecology	Population dynamics of salmon and trout, land use dynamics, flocking behaviour in fish and birds, rain forest growth
Political Sciences	Water rights in developing countries, party competition, origins and patterns of political violence, power sharing in multicultural states

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Agent-Based Simulation

Building an ABS model

1. Identify *active entities (agents)*
2. Define their states and behaviour
3. Put them in an environment
4. Establish connections
5. Test the model

Validating an ABS Model

- System behaviour is an emergent property
- Validation on a micro level
- In mixed DES/ABS it is also possible to validate on macro level



Agent-Based Simulation

Operational Research (OR)

- In OR we have combined **DES/ABS** models where we represent the process flow as a DES model and then add some active entities (to replace the passive DES entities) that are autonomous and can display proactive behaviour.

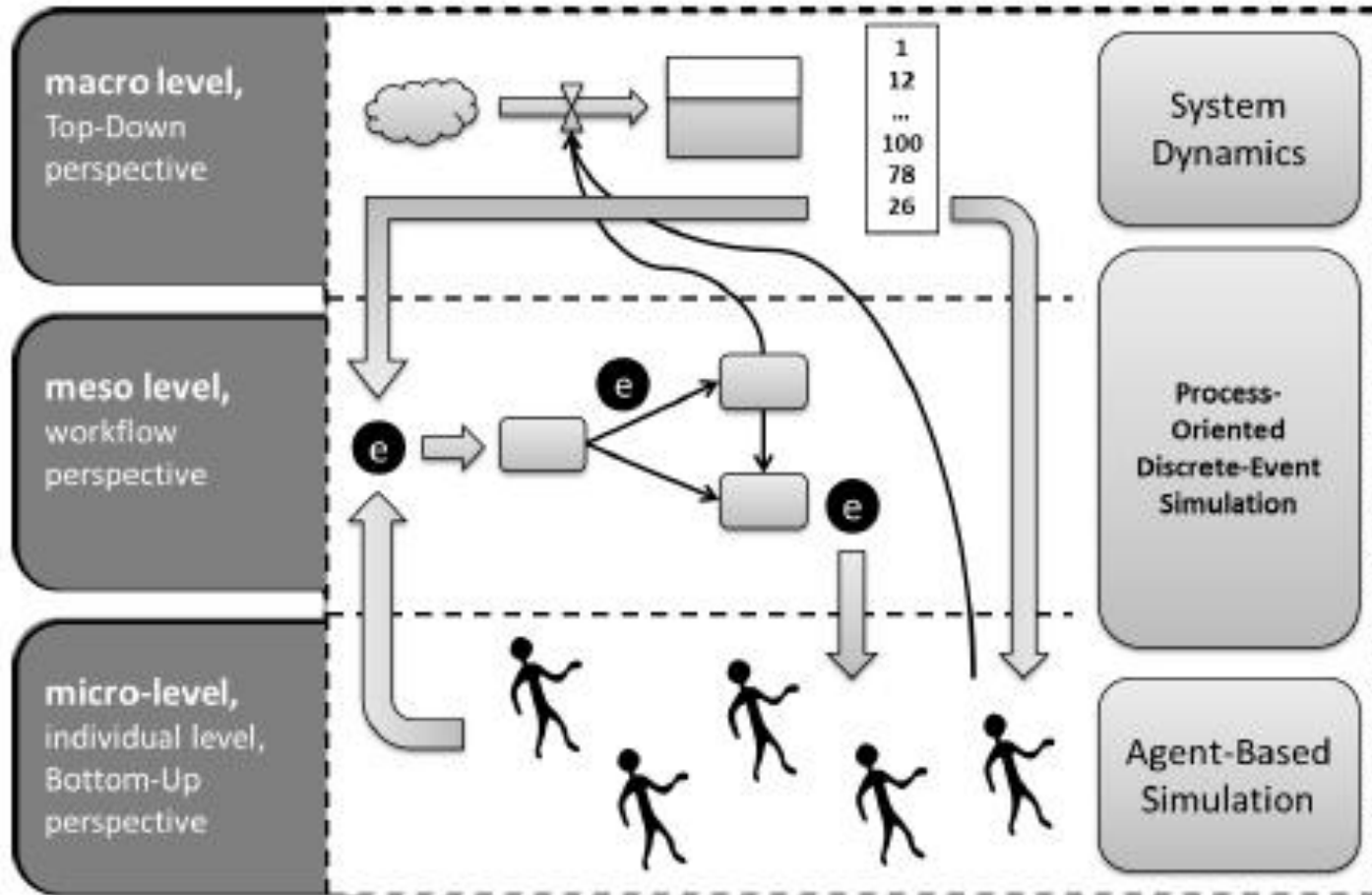
Real world applications

- Whilst many of those early examples in using ABS were concerned with general theories, there is now an increasing body of literature documenting successful applications of ABS; these examples concern traffic and transportation, financial markets, supply chains, energy usage, health and social policies

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System Dynamics + DES + Agent-Based Simulation



In the context of social analysis and systems thinking, "micro," "meso," and "macro" refer to different levels of analysis: micro focuses on individuals, meso on groups and organizations, and macro on broader systems and structures.

Figure 1: Overview of important ProHTA processes between abstraction levels.



Agent-Based Simulation - **Software**

Free:

Swarm, NetLogo, Repast, SeSAm,
GAMA...

Commercial:

AgentSheets, AnyLogic,...





Agent-Based Simulation

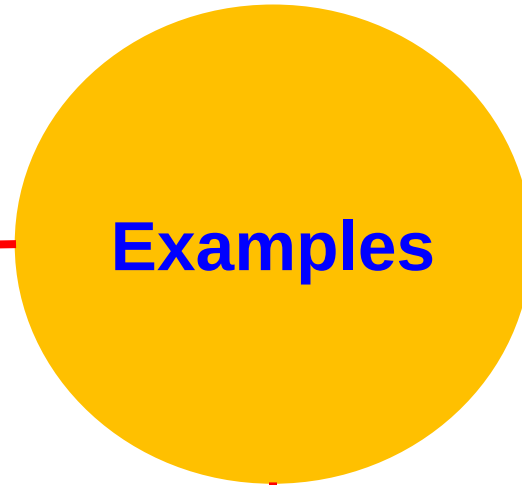


The Sims: Interactive Organisation Agent-Based Simulation

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Agent-Based Simulation - Examples



Examples

NetLogo (Biology): Ants

<https://ccl.northwestern.edu/netlogo/models/Flocking>

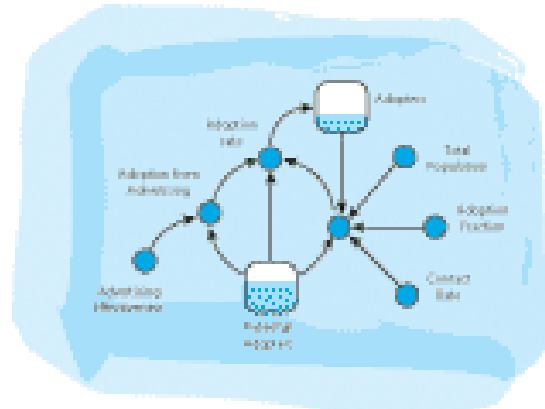
NetLogo (Social Science): Party

<https://netlogoweb.org/launch#http://netlogoweb.org/assets/modelslib/Sample%20Models/Social%20Science/Party.nlogo>

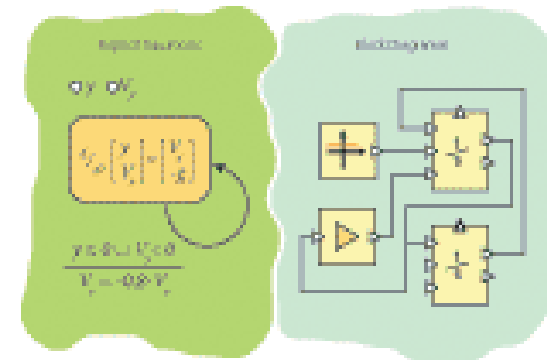
NetLogo (Social Science): Traffic Basic

<https://netlogoweb.org/launch#http://netlogoweb.org/assets/modelslib/Sample%20Models/Social%20Science/Traffic%20Basic.nlogo>

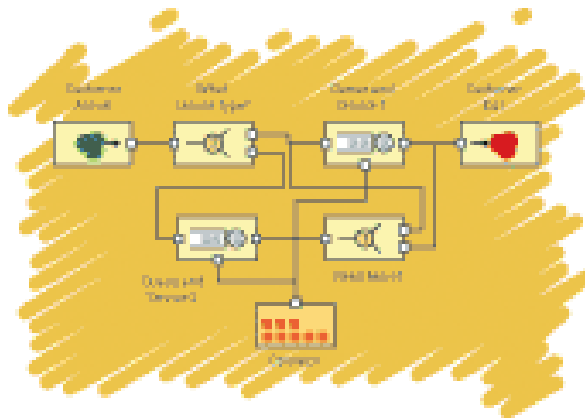
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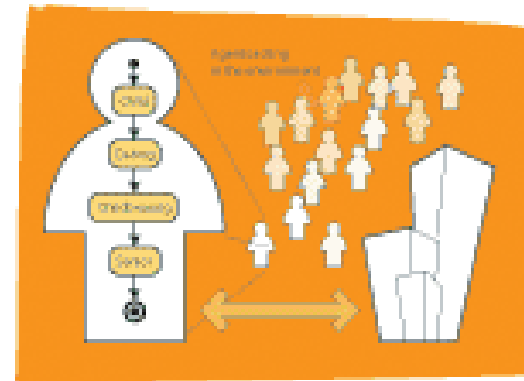
System Dynamics



Dynamic Systems



Discrete Event

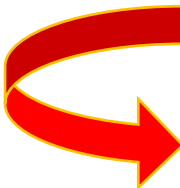


Agent Based

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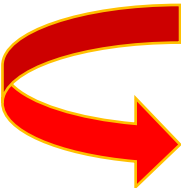


Simulation Methods



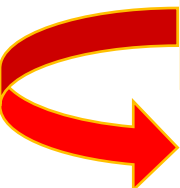
System Dynamics Simulation (continuous, deterministic)

Aggregate view; differential equations



Discrete Event Simulation (discrete, stochastic)

Process oriented (top down); one thread of control;
passive objects



Agent Based Simulation (discrete, stochastic)

Individual centric (bottom up); each agent has its own
thread of control; active objects



Simulation Methods

Feature	System Dynamics	Discrete-Event Simulations	Agent-Based Modeling
Focus	Aggregated system-level trends.	Process and queue dynamics.	Individual agent behaviors and interactions.
Granularity	High-level, aggregate.	Mid-level, event-driven.	Micro-level, individual-centric.
Time Representation	Continuous or discrete.	Discrete.	Continuous or discrete.
Modeling Style	Equation-based.	Event-based.	Rule-based.
Use Cases	Long-term trends, policy.	Operational processes.	Complex adaptive systems.

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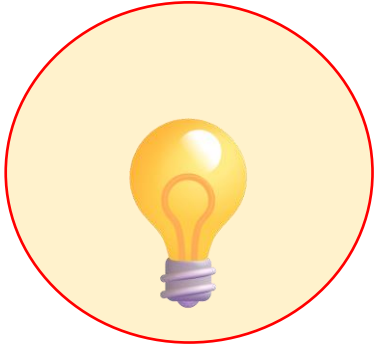
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ACK: Prof. Jie Chen, UNM; Dr. Thomas Maltz, UNM; Chen Zhiqian, UNM



Summary



1. The Nature of Simulation
2. Why Simulate?
3. Systems, Models & Simulation
4. Model Design (Conceptual Modelling)
5. Classification of Simulation Models



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Questions

NEXT:

Finale & Revision